INCH-POUND

MIL-PRF-1/61D
5 December 2006
SUPERSEDING
MIL-E-1/61C
30 March 1970

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12SK7 1/

Inactive for new design
after 21 July 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Pentode, rf remote cutoff

Outline	- - - -	8-1 (EIA)
Base	- - - -	B8-21 (phenolic)
Envelope	- - - -	MT8
Cathode	- - - -	Coated unipotential

Base connections:

Pin No.	- - - -	1	2	3	4	5	6	7	8
Element	- - - -	sh	h	g3	g1	k	g2	h	a

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Pp	Pg2	Ehk	Rg1	Alt
Unit:	V	V dc	V dc	V dc	V dc	W	W	v	MegΩ	ft
Maximum:	13.8	330	---	140	---	4.4	0.44	100	---	See note 1
Minimum:	11.4	---	---	---	---	---	---	---	---	---
Test conditions (1):	12.6	250	-3	100	0	---	---	---	---	---
Test conditions (2):	12.6	28	0	28	0	---	---	---	2	---

GENERAL:

Qualification - Not required.

1/ Formerly tube type 12SK7Y. See note 4.

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TABLE I. Testing and inspection.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 5)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part I						
1266	Total grid current	Test condition (1) (see note 2)	0.65	Ic1	0	-1.0	µA dc
1256	Electrode current (1) (anode)	Test condition (1)	0.65	Ib	6.5	12.0	mA dc
1306	Transconductance (1)	Test condition (1)	0.65	Sm	1,600	2,400	µmhos
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 30 V dc (see note 2)	0.65	Is	65	---	mA dc
1201	Short and discontinuity detection		0.4	---	---	---	---
	Conformance inspection, part 2						
1031	Low-frequency vibration	Test condition (1); Rp = 2,000 ohms	---	Ep	---	100	mV ac
1301	Heater current		---	If	138	162	mA
1336	Heater-cathode leakage		---	Ihk	---	20	µA dc
1256	Electrode current (2) (anode)	Test condition (2)	---	Ib	1.3	2.6	mA dc
1256	Electrode current (1) (screen)	Test condition (1)	---	Ic2	1.6	3.4	mA dc
1256	Electrode current (2) (screen)	Test condition (2)	---	Ic2	0.35	1.0	mA dc
1306	Transconductance (2)	Test condition (1); Ec1 = -35 V dc	---	Sm	1	30	µmhos
1306	Transconductance (3)	Test condition (2); Esig = 0.05 V (see note 3)	---	Sm	950	1,450	µmhos
1331	Direct-interelectrode capacitance	Without shield Without shield Without shield	--- 	Cg1p Cin Cout	--- 3.8 5.2	0.0035 6.2 8.8	pF pF pF
1101	Secureness of base, cap, or insert		---	---	---	---	---
1211	Insulation of electrodes		---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---
	Conformance inspection, part 3						
---	Life-test provisions	Group A; Ehk = 100 V	---	---	---	---	---
---	Life-test end point (500 hours)	Transconductance (1)	---	Sm	1,300	---	µmhos

See notes at top of next page.

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TABLE I. Testing and inspection - Continued.

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Signal coupled to grid through 1 μF capacitor.
4. Tube types 6SK7GTY and 12SK7GTY are hereby deleted from this specification sheet. For acquisition purposes, use tube type 6SK7WA for tube type 6SK7GTY and tube type 12SK7 for tube types 12SK7Y and 12SK7GTY.
5. This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-002)

Review activities:
Army - AR, CR4
Navy - AS, CG, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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